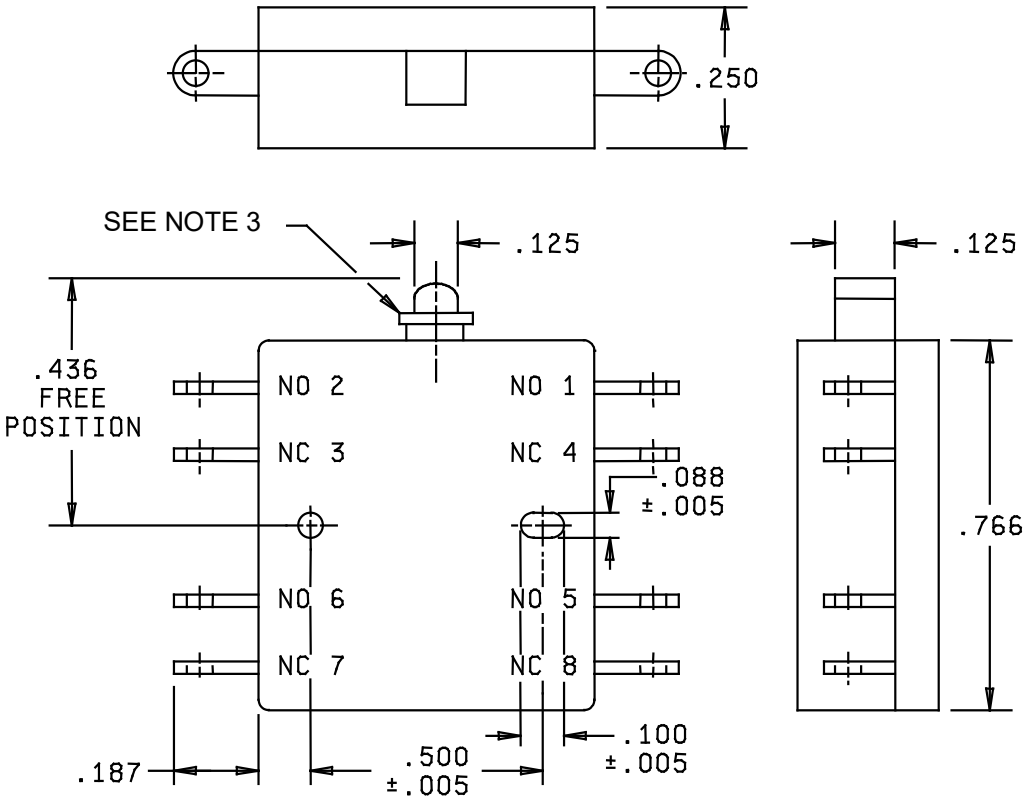
INCH-POUND
MIL-PRF-8805/77J
21 October 2019
SUPERSEDING
MIL-PRF-8805/77H
19 September, 2000

PERFORMANCE SPECIFICATION SHEET

# SWITCHES, SENSITIVE, MOMENTARY, 4 CIRCUIT OR DPDT (10 AMPERES), UNSEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-8805.



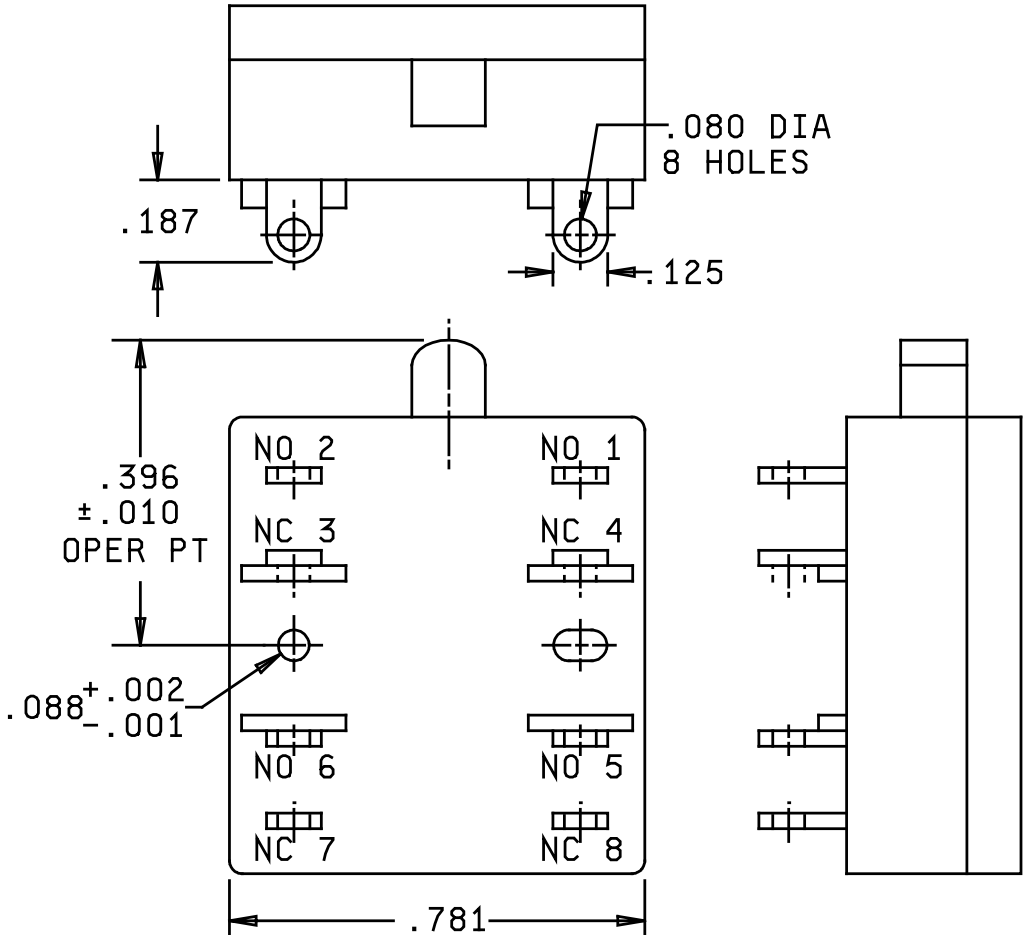

TERMINAL CONFIGURATION A

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.015.
3. Collar available on –003 and –004 only.

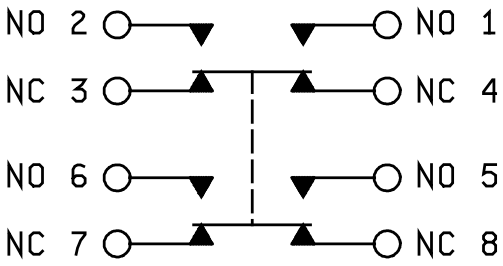
FIGURE 1. Dimensions and configuration.



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TERMINAL CONFIGURATION B




CIRCUIT SCHEMATIC

4 CIRCUIT SCHEMATIC

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 1 (unsealed) (-001 and –002).
1 (unsealed with terminal flux seal) (-003 and –004)

Temperature characteristic: 2 (-65°C to +125°C).

Shock type: M (100 g, test condition I, MIL-STD-202-213).

Vibration grade: 2 (10 Hz to 2,000 Hz swept sinusoidal, 15 g peak).

Weight: .25 ounce maximum.

Operating characteristics: ±20 percent variation from specified values acceptable after test.

Free position .436 inch.

Actuating force 1.5 pounds maximum.

Overtravel .012 inch minimum.

Movement differential .025 ±.010 inch.

Pole transfer coincidence .002 inch maximum (-003 and –004).

Operating point .396 ±.010 inch.

Terminal strength: 5 pounds, when applied perpendicular to the longitudinal axis of the terminals;
9 pounds when applied in other directions

Terminal flux sealing: M8805/77-003 and –004

Test condition:
Switches shall be tested as follows: For each switch circuit, measure and record initial contact resistance. Place switches, terminal down, into shallow pan. Pour flux, in accordance with type symbol B of J-STD-004, specific gravity 0.896, at 80°F ±5°F, into a pan without splashing until level of flux is approximately .0625 inch above the bottom of the switch case, and let switches soak for 10 minutes. Remove switches from flux and immediately place into oven for drying at 175°F ±10°F for two hours. After switches have cooled to room temperature, visually examine the actuator area for evidence of flux and repeat initial measurements. Contact resistance shall not increase by more than 10 milliohms over initial readings.

Dielectric withstanding voltage:

Sea level: 1,000 V rms initial.
1,000 V rms between all terminals and mounting plate after endurance test.
Altitude: 70,000 feet - 400 V rms.

Strength of actuator: 25 pounds.



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Logic Level circuit: 1,000,000 cycles

M8805/77-001 and -002: Not applicable
M8805/77-003 and -004: Applicable

Mechanical endurance: 100,000 cycles at .016 ±.001 inch overtravel for –001 and –002.
1,000,000 cycles at .016 ±.001 inch overtravel for –003 and –004.

Electrical endurance: 50,000 cycles for –001 and –002.
100,000 cycles for –003 and –004.

Electrical ratings: See table I and table II.

Part or Identifying Number (PIN): See table III.

QUALIFICATION:
Group submission: See table IV.

TABLE I. Electrical ratings, M8805/77-001 and -002.

| Load | Sea level | | | | 70,000 ft | |
|---|---|---|---|---|---|---|
| | 28 V dc DPDT | 28 V dc 4 circuit | 115 V ac,60 Hz DPDT | 115 V ac,60 Hz 4 circuit | 28 V dc DPDT | 28 V dc 4 circuit |
| | (amperes) | (amperes) | (amperes) | (amperes) | (amperes) | (amperes) |
| Resistive | 10 | 10 | 7.5 | 5 | 10 | 10 |
| Inductive | 7.5 | 5 | 5 | 5 | 3.5 | 2 |
| Lamp | 4 | 3 | 3 | 1.5 | --- | --- |

TABLE II. Electrical ratings, M8805/77-003 and -004.

| Load | Sea level | | | | 70,000 ft | | Logic level |
|---|---|---|---|---|---|---|---|
| | 28 V dc DPDT | 28 V dc 4 circuit | 115 V ac, 60 Hz DPDT | 115 V ac, 60 Hz 4 circuit | 28 V dc DPDT | 28 V dc 4 circuit | 5 V dc |
| | amperes | Amperes | amperes | amperes | amperes | amperes | milliamperes |
| Resistive | 8 | 8 | 6 | 4 | 8 | 8 | 10 |
| Inductive | 6 | 4 | 4 | 4 | 2 | 1 | --- |
| Lamp | 0.5 | 0.5 | 2 | 1 | --- | --- | --- |

TABLE III. PIN and terminal configurations.

| PIN | Terminal configuration |
|---|---|
| M8805/77-001 and –003<br>M8805/77-002 and –004 | A (end solder type)<br>B (side solder type) |



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TABLE IV. Group submission.

| PIN | Examination or test | Extent of approval |
|---|---|---|
| M8805/77-001 and –003 (all sample units) | Qualification inspection table of MIL-PRF-8805 | All |
| M8805/77-002 and –004 (3 sample units) | Terminal strength | |
| | Thermal shock | |
| | Operating characteristics | |
| | Visual and mechanical examination | |
| | Flux seal | |

Referenced documents:
MIL-PRF-8805 MIL-STD-202-213 J-STD-004

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2019-057)

Review activities:
Army - AT, AV, CR4

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .



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